BO

- Acquisition functions in BO
- Examples

BO loops

We consider an optimization problem

$$\min_{x \in \mathcal{X}} f(x) \qquad \text{or} \qquad \max_{x \in \mathcal{X}} f(x)$$

with a probabilistic surrogate

$$g \sim \mathrm{GP}_{\mathcal{X}}(m, C)$$
$$y_n | g \sim \mathrm{Normal}\left(g(x_n), \sigma^2\right), \qquad n = 1, \ldots, N$$

for appropriate m, C and σ

BO starts with: *...this is the initialization stage*

- a pre-selected set of queries
- a pre-evaluated set of targets at the queries

...iteration $j = 0$

BO proceeds iteratively: *...this is the main loop*

- form posterior surrogate using existing data
- apply the surrogate to select new query
- evaluate new target
- update data

...iterations $j = 1, 2, \ldots$

Acquisition functions

Within our BO loop, at iteration $j = 1, 2, \ldots$ we have data and posterior surrogate

$$\{(x_n, y_n)\}_{n=1}^{N^{j-1}} \subset \mathcal{X} \times \mathbb{R}$$
$$g(x)|y_{1:N^{j-1}} \sim \text{Normal}\left(m(x; y_{1:N^{j-1}}), v(x; y_{1:N^{j-1}})\right)$$

To balance exploitation/exploration across the search space, we derive acquisition functions

$$\alpha^j : \mathcal{X} \mapsto \mathbb{R}$$

We cast the selection of the next query point as an optimization

$$x_{N^j} = \text{argmax}_{x \in \mathcal{X}} \alpha^j(x) \qquad x_{N^j} = \text{argmin}_{x \in \mathcal{X}} \alpha^j(x)$$
$$y_{N^j} = f(x_{N^j}) \qquad y_{N^j} = f(x_{N^j})$$

since α^j models expected utility or expected loss under posterior uncertainty

UCB/LCB

Within our BO loop, at iteration $j = 1, 2, \ldots$ we have data and posterior surrogate

$$\{(x_n, y_n)\}_{n=1}^{N^{j-1}} \subset \mathcal{X} \times \mathbb{R}$$
$$g(x)|y_{1:N^{j-1}} \sim \text{Normal}\left(m(x; y_{1:N^{j-1}}), v(x; y_{1:N^{j-1}})\right)$$

- For maximization problems, $\max_{x \in \mathcal{X}} f(x)$, the upper confidence bound is

$$\alpha^j(x) = m(x) + \beta\sqrt{v(x)}$$

and the next query is selected by

$$x_{Nj} = \text{argmax}_{x \in \mathcal{X}} \alpha^j(x)$$

- For minimization problems, $\min_{x \in \mathcal{X}} f(x)$, the lower confidence bound is

$$\alpha^j(x) = m(x) - \beta\sqrt{v(x)}$$

and the next query is selected by

$$x_{Nj} = \text{argmin}_{x \in \mathcal{X}} \alpha^j(x)$$

In both cases, $\beta > 0$ controls the exploitation/exploration trade-off

...with $\beta \ll 1$ favoring exploitation
...with $\beta \gg 1$ favoring exploration

EI

Within our BO loop, at iteration $j = 1, 2, \ldots$ we have data and posterior surrogate

$$\{(x_n, y_n)\}_{n=1}^{N^{j-1}} \subset \mathcal{X} \times \mathbb{R}$$
$$g(x)|y_{1:N^{j-1}} \sim \text{Normal}\left(m(x; y_{1:N^{j-1}}), v(x; y_{1:N^{j-1}})\right)$$

- For maximization problems, $\max_{x \in \mathcal{X}} f(x)$, the expected improvement is

$$y_*^j = \max y_{1:N^{j-1}}$$
$$\alpha^j(x) = \left(\mu(x) - y_*^j\right) \Phi\left(\frac{\mu(x) - y_*^j}{\sqrt{v(x)}}\right) + \sqrt{v(x)}\phi\left(\frac{\mu(x) - y_*^j}{\sqrt{v(x)}}\right)$$

and the next query is selected by

$$x_{N^j} = \text{argmax}_{x \in \mathcal{X}} \alpha^j(x)$$

- For maximization problems, $\min_{x \in \mathcal{X}} f(x)$, the expected improvement is

$$y_*^j = \min y_{1:N^{j-1}}$$
$$\alpha^j(x) = \left(y_*^j - \mu(x)\right) \Phi\left(\frac{y_*^j - \mu(x)}{\sqrt{v(x)}}\right) + \sqrt{v(x)}\phi\left(\frac{y_*^j - \mu(x)}{\sqrt{v(x)}}\right)$$

and the next query is selected by

$$x_{N^j} = \text{argmax}_{x \in \mathcal{X}} \alpha^j(x)$$

In both cases, $\Phi(z)$ and $\phi(z)$ denote the CDF and PDF of the standard normal

EI with jitter

Within our BO loop, at iteration $j = 1, 2, \ldots$ we have data and posterior surrogate

$$\{(x_n, y_n)\}_{n=1}^{N^{j-1}} \subset \mathcal{X} \times \mathbb{R}$$
$$g(x)|y_{1:N^{j-1}} \sim \text{Normal}\left(m(x; y_{1:N^{j-1}}), v(x; y_{1:N^{j-1}})\right)$$

- For maximization problems, $\max_{x \in \mathcal{X}} f(x)$, the expected improvement is

$$y_*^j = \max y_{1:N^{j-1}} + \xi$$
$$\alpha^j(x) = \left(\mu(x) - y_*^j\right) \Phi\left(\frac{\mu(x) - y_*^j}{\sqrt{v(x)}}\right) + \sqrt{v(x)}\phi\left(\frac{\mu(x) - y_*^j}{\sqrt{v(x)}}\right)$$

and the next query is selected by

$$x_{N^j} = \text{argmax}_{x \in \mathcal{X}} \alpha^j(x)$$

In both cases, $\xi > 0$ encourages more exploration

- For maximization problems, $\min_{x \in \mathcal{X}} f(x)$, the expected improvement is

$$y_*^j = \min y_{1:N^{j-1}} - \xi$$
$$\alpha^j(x) = \left(y_*^j - \mu(x)\right) \Phi\left(\frac{y_*^j - \mu(x)}{\sqrt{v(x)}}\right) + \sqrt{v(x)}\phi\left(\frac{y_*^j - \mu(x)}{\sqrt{v(x)}}\right)$$

and the next query is selected by

$$x_{N^j} = \text{argmax}_{x \in \mathcal{X}} \alpha^j(x)$$

PI with jitter

Within our BO loop, at iteration $j = 1, 2, \ldots$ we have data and posterior surrogate

$$\{(x_n, y_n)\}_{n=1}^{N^{j-1}} \subset \mathcal{X} \times \mathbb{R}$$
$$g(x)|y_{1:N^{j-1}} \sim \text{Normal}\left(m(x; y_{1:N^{j-1}}), v(x; y_{1:N^{j-1}})\right)$$

- For maximization problems, $\max_{x \in \mathcal{X}} f(x)$, the probability of improvement is

$$y_*^j = \max y_{1:N^{j-1}} + \xi$$
$$\alpha^j(x) = \Phi\left(\frac{\mu(x) - y_*^j}{\sqrt{v(x)}}\right)$$

and the next query is selected by

$$x_{N^j} = \text{argmax}_{x \in \mathcal{X}} \alpha^j(x)$$

In both cases, $\xi \geq 0$ encourages more exploration

- For maximization problems, $\min_{x \in \mathcal{X}} f(x)$, the probability of improvement is

$$y_*^j = \min y_{1:N^{j-1}} - \xi$$
$$\alpha^j(x) = \Phi\left(\frac{y_*^j - \mu(x)}{\sqrt{v(x)}}\right)$$

and the next query is selected by

$$x_{N^j} = \text{argmax}_{x \in \mathcal{X}} \alpha^j(x)$$

TS

Within our BO loop, at iteration $j = 1, 2, \ldots$ we have data and posterior surrogate

$$\{(x_n, y_n)\}_{n=1}^{N^{j-1}} \subset \mathcal{X} \times \mathbb{R}$$
$$g(x)|y_{1:N^{j-1}} \sim \text{Normal}\left(m(x; y_{1:N^{j-1}}), v(x; y_{1:N^{j-1}})\right)$$

- For maximization problems, $\max_{x \in \mathcal{X}} f(x)$, Thompson sampling proceeds stochastically

$$\alpha^j = g|y_{1:N^{j-1}}$$

and the next query is selected by

$$x_{N^j} = \text{argmax}_{x \in \mathcal{X}} \alpha^j(x)$$

- For maximization problems, $\min_{x \in \mathcal{X}} f(x)$, Thompson sampling proceeds stochastically

$$\alpha^j = g|y_{1:N^{j-1}}$$

and the next query is selected by

$$x_{N^j} = \text{argmax}_{x \in \mathcal{X}} \alpha^j(x)$$

In both cases, the acquisition function is our posterior surrogate, which is generated stochastically in each iteration

Myopic and non-myopic policies

Myopic policies use acquisition functions are those that consider only one-step ahead

...ie, evaluate the immediate benefit of acquiring a query assuming this is the final round

...typically come with closed-formulas

Non-myopic policies use acquisition functions are those that plan multiple-steps ahead

...ie, evaluate the immediate benefit of acquiring a query as well as its improvement on subsequent evaluations

...typically do not come with closed-formulas

Non-myopic acquisitions simulate a sequence of future queries and consider how each sequence reduces uncertainty while improving subsequent decisions across the remaining budget

...they are computationally expensive

...they rely on approximations

Demonstration

We seek to maximize an expensive objective on a 1D search space

$$f : [x_{min}, x_{max}] \mapsto \mathbb{R}, \qquad x_{min} = -5, \qquad x_{max} = +10$$

- Our surrogate is determined by

$$g \sim \mathrm{GP}_{\mathcal{X}}(m, C)$$

$$y_n | g \sim \mathrm{Normal}\left(g(x_n), \sigma^2\right), \qquad n = 1, \ldots$$

with hyper-settings given by

$$m(x) = \zeta, \qquad \zeta = 20$$

$$C(x', x'') = \gamma^2 \exp\left(-\left(\frac{x' - x''}{\ell}\right)^2\right), \qquad \gamma = 10, \qquad \ell = 1$$

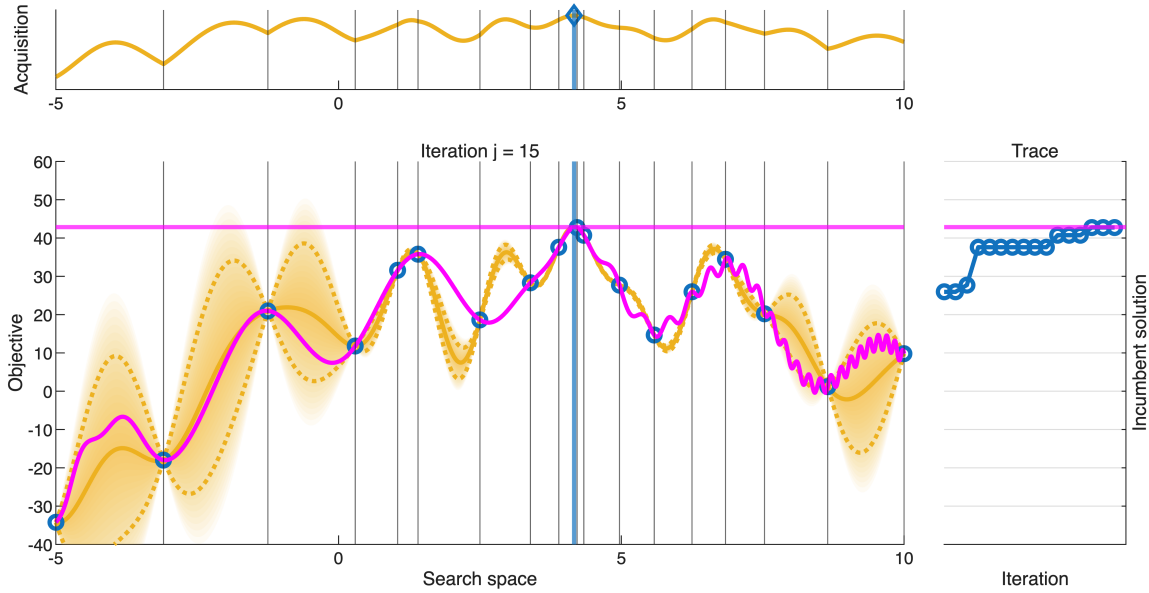
$$\sigma \ll 1$$

- Our initial design is determined by

$$x_n = x_{min} + n\frac{x_{max} - x_{min}}{N^0 + 1}, \qquad n = 1, \ldots, N^0, \qquad N^0 = 3$$

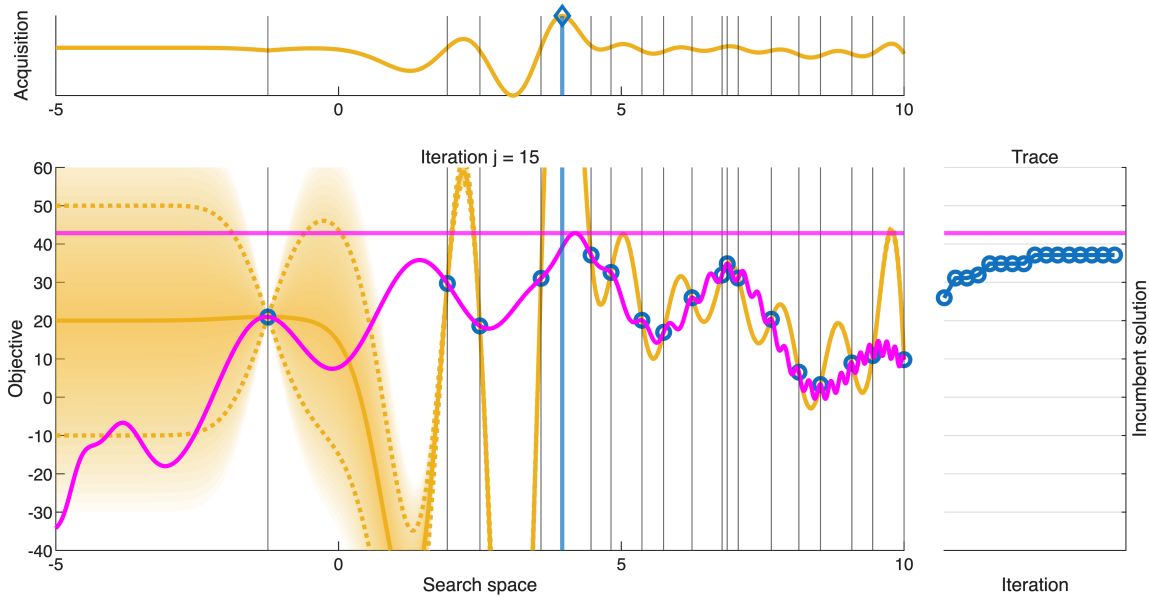
UCB

$\beta = 3$


Acquisition
-5
0
5
10
Iteration j = 15
Objective
60
50
40
30
20
10
0
-10
-20
-30
-40
-5
0
5
10
Search space
Trace
Incumbent solution
Iteration

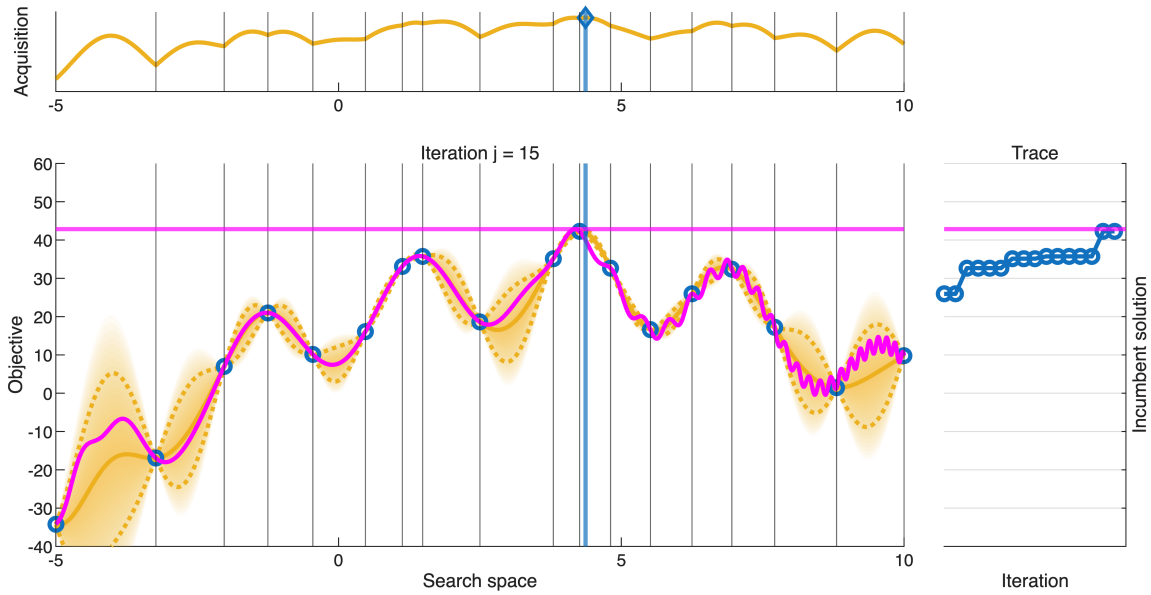
UCB

$\beta = 1$

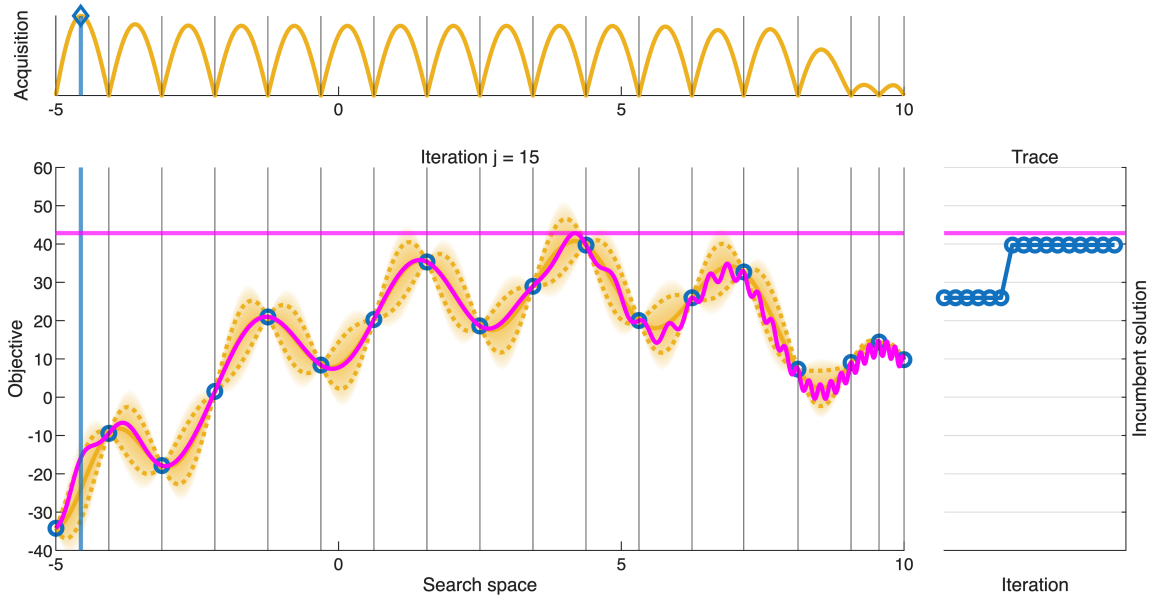

Acquisition
-5
0
5
10
Iteration j = 15
Trace
60
50
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0
-10
-20
-30
-40
Objective
Incumbent solution
Search space
Iteration

UCB

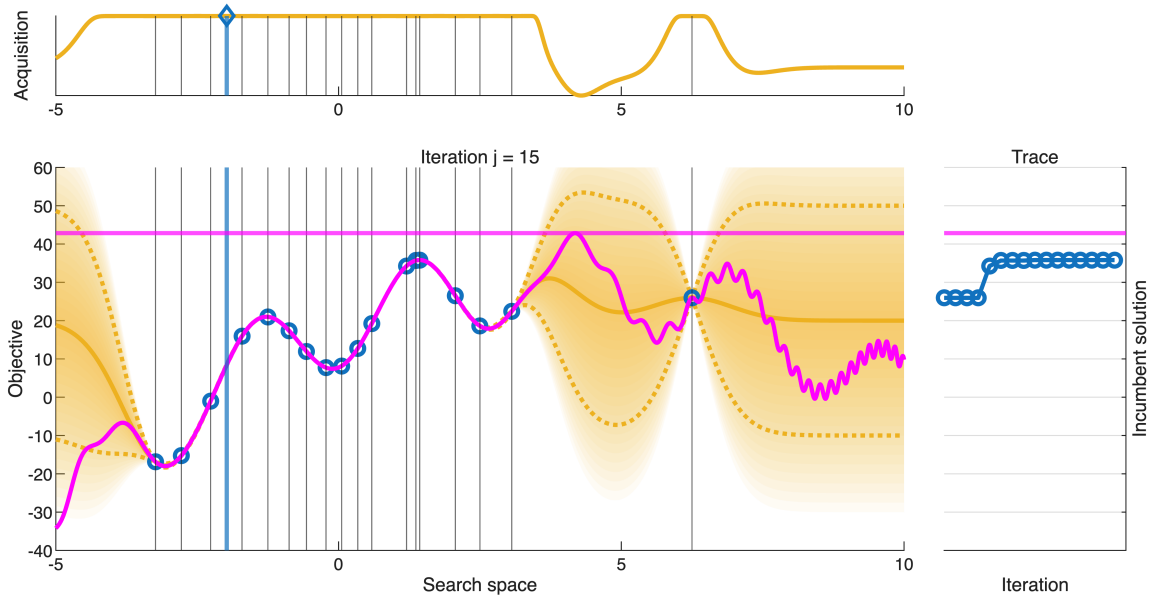
$\beta = 5$


Acquisition
-5
0
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Iteration j = 15
Objective
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-10
-20
-30
-40
Search space
Trace
Incumbent solution
Iteration

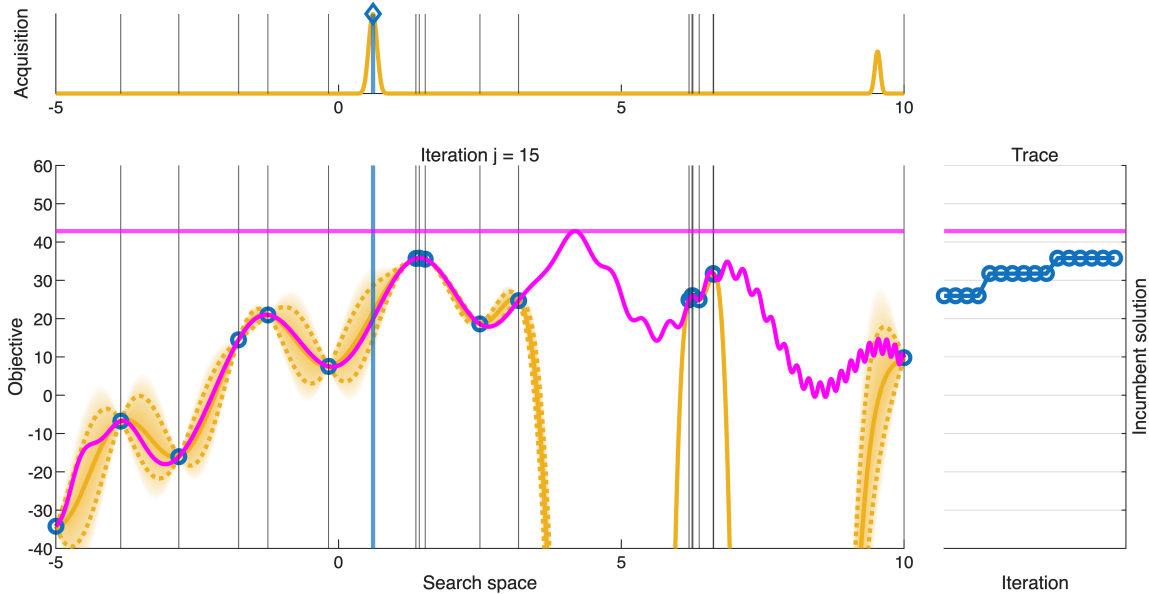
US


Acquisition
-5
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Iteration j = 15
Objective
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-10
-20
-30
-40
-5
0
5
10
Search space
Trace
Incumbent solution
Iteration

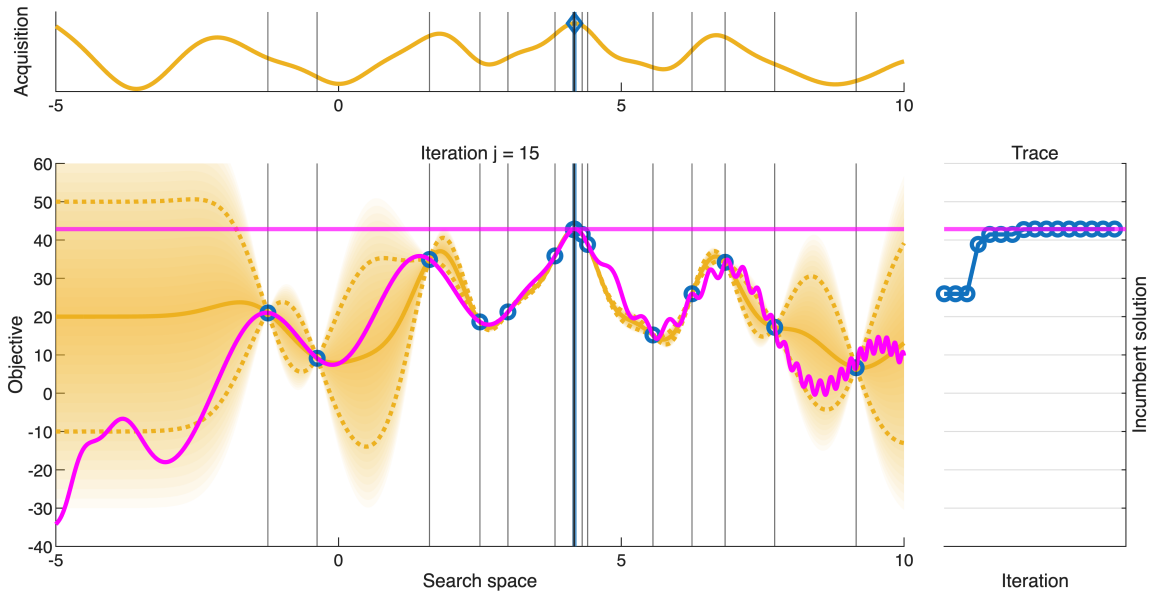
EI


Acquisition
-5
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Iteration j = 15
Objective
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-10
-20
-30
-40
Search space
Trace
Incumbent solution
Iteration

PI


Acquisition
-5
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Iteration j = 15
Trace
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-10
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-30
-40
Objective
Incumbent solution
Search space
Iteration

TS


Acquisition
-5
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Iteration j = 15
Objective
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Search space
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Iteration